Tahtabir Rahman

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Professional Summary

Data Engineer with **4+ years** of experience building reliable **ETL pipelines** that turn raw data from multiple sources into validated, analytics-ready datasets across the **supply chain and banking** domains. Currently building an end-to-end **AWS data platform** for global rubber supply chain data using **Python, PySpark, SQL, AWS Glue, S3, Athena, Lambda, and Step Functions**, with star-schema data models and orchestrated, monitored workflows. Earlier, at HCL Tech for **Barclays**, automated ServiceNow and operational data pipelines with Python (Pandas) and SQL, cutting manual data handling by **30%**. Strong in data quality, reconciliation and performance tuning. Partners with engineering and business teams to deliver dependable data for reporting, BI and compliance.

Technical Skills

Languages: Python (Pandas, NumPy, SQLAlchemy), SQL
Big Data & Processing: PySpark, Apache Spark, Batch & Incremental Processing
AWS Cloud: S3, Glue, Glue Data Catalog, Athena, Lambda, Step Functions, CloudWatch, IAM
Data Engineering: ETL Pipelines, Data Lake (Raw/Staging/Curated), Metadata-Driven Pipelines, Pipeline Orchestration & Automation
Data Modeling & SQL: Star Schema (Fact/Dimension), Joins, CTEs, Window Functions, Partitioning, Bucketing, Query Optimization
Data Quality: Validation, Reconciliation, Deduplication, Schema Enforcement, Data Profiling, Exception Handling
Databases & Formats: MySQL, DynamoDB, Parquet, CSV, JSON
Tools & DevOps: Git, GitHub, CI/CD, ServiceNow
BI & Reporting: Power BI, Tableau, Advanced Excel
AI-Assisted Development: ChatGPT, Claude (code generation, debugging, SQL optimization, test cases, documentation)

Experience

Data Engineer India
AsmaWebs Pvt Ltd. | Client: Halcyon Agri Corporation Limited | Domain: Supply Chain *Oct 2025 – Present*
Project: Global Rubber Supply Chain Data Platform
Tech Stack: Python, PySpark, SQL, AWS (S3, Glue, Athena, Step Functions, Lambda, CloudWatch)

- **Description:** Halcyon Agri is a global supplier of natural rubber with operations in several countries. The project built an automated, end-to-end data platform that ingests, cleans and models large volumes of supply chain and inventory data from international operations. The ETL flow ingests raw data from REST APIs and staging servers into an AWS S3 landing zone, then cleans and aggregates it in PySpark. Validation and business rules run next, and the data is modeled into star-schema tables in Athena, which reporting uses as one consistent source.
- Built PySpark batch pipelines on AWS Glue that clean, aggregate and transform multi-gigabyte supply chain and inventory logs from international operations.
- Automated daily ingestion from REST APIs and staging servers into an S3 raw landing zone. Orchestration ran through Step Functions and Lambda, with retry logic and CloudWatch monitoring.
- Designed star-schema data models (fact and dimension tables) and created Athena table schemas, so reporting and analytics had analytics-ready datasets.
- Built Python and SQL data quality checks that catch schema drift, null values and duplicate records before data reaches downstream reports.
- Tuned complex SQL joins, CTEs, partitioning and file layout to shorten daily pipeline runs and cut Athena query latency.

Analyst Noida, India
HCL Tech | Client: Barclays | Domain: Banking & Financial Services *Apr 2022 – Oct 2025*
Project: IT Operations Data Pipelines & Reporting
Tech Stack: Python (Pandas), SQL, Data Modeling, AWS S3, ServiceNow

- **Description:** Barclays is a major global bank offering retail, corporate and investment banking services. This project automated data extraction, transformation and reporting for Barclays' IT operations. Data came from ServiceNow and other operational sources and was consolidated into analytics-ready datasets for the networking,

cybersecurity and IT operations teams. The ETL flow extracted raw service-management data, cleansed and transformed it with Python and SQL, and applied validation and reconciliation rules. The processed datasets were stored in AWS S3 for recurring operational reporting and regulatory compliance.

- Built automated Python and SQL pipelines that replaced manual data preparation, cutting manual data handling by **30%** and making processing more consistent.
- Developed ETL workflows using Python (Pandas) and SQL that extract data from ServiceNow and operational sources, then cleanse, transform, validate and consolidate it into analytics-ready datasets.
- Designed reusable SQL data models (tables and views) that structure operational and service-management data for recurring reporting and banking compliance requirements.
- Added Pandas- and SQL-based validation and reconciliation checks that catch missing, duplicate and inconsistent records across operational and compliance datasets, and fixed discrepancies through root-cause analysis.
- Optimized SQL queries and Python scripts and organized data in AWS S3 for structured storage and downstream use, so datasets were delivered within SLA.

Education

Maulana Azad College of Engineering & Technology, Patna 2017 – 2021
Bachelor of Technology in Computer Science and Engineering

Certifications

AWS Cloud Practitioner Bootcamp (HCL Tech) Link
Power BI (PL-300) (HCL Tech) Link
Foundations: Data, Data, Everywhere (Coursera) Link
Ask Questions to Make Data-Driven Decisions (Coursera) Link
Machine Learning (Coursera) Link
Microsoft Excel (Coursera) Link

Extra Contributions

Volunteering: Received multiple appreciations from organizations for exceeding project expectations (2022–2025)